Reactive H_2S Stripping Column Simulation Software with CO_2 Hydration Kinetics

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Abstract

Traditional equilibrium-based models fail to predict reactive stripping performance by neglecting CO_2 hydration kinetics. This study develops a stage-wise Python model for converting NaHS to $NaHCO_3$ that explicitly incorporates slow CO_2 hydration kinetics ($k_f = 0.030259\ s^{-1}$ at 25°C). The model integrates mass balances, chemical equilibria, and temperature-dependent properties using scipy.optimize.fsolve to solve nonlinear equation systems per stage. Sensitivity analyses reveal that kinetic effects dominate at high H_2S recovery targets (>97%), with optimal conditions at 20°C and 2.8 atm minimizing stage requirements. Results demonstrate significant deviations from equilibrium predictions, particularly at recovery targets above 99%, where actual H_2S partial pressures exceed theoretical values by up to 19%. The validated model provides accurate process design tools for industrial sodium carbonate production via reactive stripping

Keywords: Reactive stripping column; mass transfer; non-equilibrium processes; stage-wise modelling; hydrogen sulphide removal.

Required Metadata

Table 1: Code Metadata

Nr	Code Metadata Description	Code Metadata
C1	Current code version	v2.0.1
C2	Permanent link to code/repository used of this code version	Repository: https://github.com/DamianvanTonder/Reactive_H2S_Stripping_Column_Simulation_Model_with_CO2_Hydration_Kinetics
C3	Code Ocean compute capsule	N/A
C4	Legal Code License	MIT Licence
C5	Code versioning system used	git
C6	Software code languages, tools, and services used	• Python 3.13.3 • NumPy 2.2.5 • SciPy 1.15.3 • Pandas 2.2.3 • Matplotlib 3.10.3
C7	Compilation requirements, operating environments & dependencies	• Python ≥ 3.8.0 • NumPy ≥ 1.20.0 • SciPy ≥ 1.7.0 • Pandas ≥ 1.3.0 • Matplotlib ≥ 3.3.0
C8	If available Link to developer documentation/manual	Developer Documentation: https://github.com/DamianvanTonder/Reactive_H2S_Stripping_Column_Simulation_Model_with_CO2_Hydration_Kinetics/tree/CSC421_Research_Project/docs
C9	Support email for questions	u21449300@tuks.co.za

1. Motivation and Significance

The global chemical industry is continually pursuing sustainable routes to convert low-value industrial byproducts into high-value products. Sodium sulphate (Na_2SO_4), a common waste from pulp and paper, textile, and chemical industries, can be transformed into sodium carbonate (Na_2CO_3)—a valuable compound used extensively in glass manufacturing and detergent production [1]. This conversion can be achieved through a reactive stripping process, where hydrogen sulphide (H_2S) is stripped from an aqueous sodium hydrosulphide (NaHS) solution using carbon dioxide (CO_2). The reaction produces sodium bicarbonate ($NaHCO_3$), which is then thermally decomposed to yield sodium carbonate, creating a circular process that transforms industrial waste into a commercially valuable product.

The major challenge in developing this process lies in accurately modelling the coupled phenomena of chemical equilibria, reaction kinetics, and gas-liquid mass transfer. Traditional equilibrium-based models assume that all reactions occur instantaneously, which leads to significant errors in predicting real system behaviour. In reality, the hydration of CO_2 in water to form carbonic acid (H_2CO_3) is considerably slower than the subsequent dissociation steps and has been identified as the rate-limiting reaction in similar aqueous CO_2 systems. Dreybrodt [2] first highlighted this kinetic limitation in H_2O–CO_2–$CaCO_3$ systems, and later work by Soli and Byrne [3] quantified the forward hydration rate constant as only 0.030259 s^{-1} at 25°C. Neglecting this kinetic bottleneck results in substantial deviations between model predictions and experimental data—particularly under conditions where H_2S recovery is high and the chemical driving forces are weak.

Accurate representation of reactive stripping also depends heavily on mass transfer characteristics and physical property correlations. Studies by Nock [4] and Guo [5] established relationships for CO_2 bubble dynamics and rise velocities in aqueous systems, while Perry's Chemical Engineering Handbook [6] provides the necessary correlations for liquid viscosity, density, and surface tension. Thermodynamic data for the dissociation of carbonic acid and hydrogen sulphide were provided by Millero [7,8] and Stephens and Cobble [9], while water ionization constants across temperature ranges were determined by Olofsson and Hepler [10]. Together, these data form the foundation for an accurate thermodynamic and kinetic description of the process.

This research addresses a critical modelling gap by developing a computational framework that explicitly incorporates CO_2 hydration kinetics alongside mass transfer and equilibrium phenomena in multistage reactive stripping columns. The software, implemented in Python, allows users to input key process parameters (temperature, pressure, flow rates, and feed compositions) and obtain detailed predictions of column performance, including pH profiles, species distributions, and gas compositions. Automated sensitivity analyses identify optimal operating conditions and highlight the regimes where kinetic effects dominate.

The developed model provides a rigorous, physically grounded tool for the design and optimization of reactive stripping processes, supporting future experimental validation and industrial application in sustainable Na_2CO_3 production from waste Na_2SO_4.

2. Materials and Methodology

2.1. Overview and Software Framework

The investigation was conducted entirely through computational modelling using Python 3.13.3, with no experimental components. Figure 1 presents a schematic of the reactive stripping column configuration used in the model.

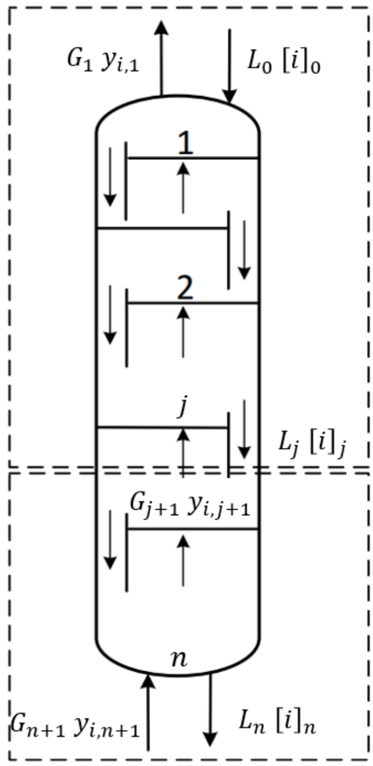

Figure 1: Schematic Drawing of the Reactive Stripping Column

In Figure 1, the symbols L_j and G_j indicate respectively liquid and vapour streams flowing from stage j in the column, while $[i]_j$ and $y_{i,j}$ indicate concentrations of species i in the respective liquid or vapour stream flowing from stage j.

The computational framework utilized NumPy 2.2.5 for array operations and mathematical functions, SciPy 1.15.3 for nonlinear equation solving via optimize.fsolve employing the hybrid Powell method, Pandas 2.2.3 for data manipulation and organizing stage-wise results into structured DataFrames, and Matplotlib 3.10.3 for visualization of concentration profiles, pH trends, and convergence diagnostics. Development occurred in Jupyter Notebook to facilitate iterative refinement and real-time debugging of the complex equation systems.

2.2. Mathematical Model Structure

2.2.1. Reactive Stripping Column Model

The stage-wise model represents each stage as a theoretical equilibrium stage modified by kinetic limitations. Six chemical reactions were incorporated:

$$CO_2(aq) + H_2O(l) \rightleftharpoons H_2CO_3(aq)$$

$$H_2CO_3(aq) \rightleftharpoons H^+(aq) + HCO_3^-(aq)$$

$$HCO_3^-(aq) \rightleftharpoons H^+(aq) + CO_3^{2-}(aq)$$

$$H_2S(aq) \rightleftharpoons H^+(aq) + HS^-(aq)$$

$$HS^-(aq) \rightleftharpoons H^+(aq) + S^{2-}(aq)$$

$$H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$$

The overall reaction occurring in the reactive stripping column is:

$$NaHS(aq) + CO_2(g) + H_2O(l) \rightleftharpoons NaHCO_3(aq) + H_2S(g)$$

The following equations and balances were derived to solve for the reactive stripping column, starting at the first stage at the top of the column.

For the first stage of the column only, the overall H_2S recovery for the column which predicts the amount of H_2S leaving the column in the vapor stream was defined as:

$$G_1 y_{H_2S,1} = L\,[NaHS]_0\, x_{H_2S} \tag{1}$$

The CO_2 balance was derived:

$$\begin{aligned} G_{j+1}\, y_{CO_2,j+1} + L\big([CO_2(aq)]_{j-1} + [H_2CO_3]_{j-1} + [HCO_3^-]_{j-1} + [CO_3^{2-}]_{j-1}\big) \\ = G_j\, y_{CO_2,j} + L\big([CO_2(aq)]_j + [H_2CO_3]_j + [HCO_3^-]_j + [CO_3^{2-}]_j\big) \end{aligned} \tag{2}$$

The sulphur balance was derived:

$$\begin{aligned} G_{j+1}\, y_{H_2S,j+1} + L\big([H_2S(aq)]_{j-1} + [HS^-]_{j-1} + [S^{2-}]_{j-1}\big) \\ = G_j\, y_{H_2S,j} + L\big([H_2S(aq)]_j + [HS^-]_j + [S^{2-}]_j\big) \end{aligned} \tag{3}$$

The charge balance was derived:

$$[H^+]_j + [Na^+] = [OH^-]_j + [HCO_3^-]_j + 2[CO_3^{2-}]_j + [HS^-]_j + 2[S^{2-}]_j \tag{4}$$

The CO_2 reaction rate was derived:

$$\begin{aligned} k_f\left([CO_2(aq)]_j - \frac{[H_2CO_3]_j}{K_{eq}}\right)V(1-\varepsilon) = L\big([H_2CO_3]_j + [HCO_3^-]_j + [CO_3^{2-}]_j\big) \\ - L\big([H_2CO_3]_{j-1} + [HCO_3^-]_{j-1} + [CO_3^{2-}]_{j-1}\big) \end{aligned} \tag{5}$$

For any number of vapor mole fractions:

$$y_{CO_2,j+1} + y_{H_2S,j+1} + y_{H_2O} = 1 \tag{6}$$

The mass transfer of CO_2 was defined as:

$$k_{L,CO_2}\, a\, V\big([CO_2(i)]_j - [CO_2(aq)]_j\big) = G_{j+1}\, y_{CO_2,j+1} - G_j\, y_{CO_2,j} \tag{7}$$

Similarly, the mass transfer of H_2S in water was defined as:

$$k_{L,H_2S} a\, V\left([H_2S(i)]_j - [H_2S(aq)]_j\right) = G_{j+1}\, y_{H_2S,j+1} - G_j\, y_{H_2S,j} \tag{8}$$

The two mass transfer equations was added to define:

$$G_{j+1} = G_j + \frac{k_{L,CO_2} a\, V\left([CO_2(i)]_j - [CO_2(aq)]_j\right) + k_{L,H_2S} a\, V\left([H_2S(i)]_j - [H_2S(aq)]_j\right)}{1 - y_{H_2O}} \tag{9}$$

The equilibrium expression for the first dissociation of carbonic acid was defined as:

$$K_{1C} = \frac{[H^+]_j\, [HCO_3^-]_j}{[H_2CO_3]_j} \tag{10}$$

The equilibrium expression for the second dissociation of carbonic acid was defined as:

$$K_{2C} = \frac{[H^+]_j\, [CO_3^{2-}]_j}{[HCO_3^-]_j} \tag{11}$$

The equilibrium expression for the first dissociation of hydrogen sulphide was defined as:

$$K_{1S} = \frac{[H^+]_j\, [HS^-]_j}{[H_2S(aq)]_j} \tag{12}$$

The equilibrium expression for the second dissociation of hydrogen sulphide was defined as:

$$K_{2S} = \frac{[H^+]_j\, [S^{2-}]_j}{[HS^-]_j} \tag{13}$$

The equilibrium expression for the autoionization of water was defined as:

$$K_W = [H^+]_j\, [OH^-]_j \tag{14}$$

The equilibrium between the CO_2 gas and the dissolved CO_2 at the gas-liquid interface was defined as:

$$[CO_2(i)]_j = k'_{H,CO_2}\, P_{CO_2} = k'_{H,CO_2}\, y_{CO_2,j}\, P_{tot} \tag{15}$$

For the equilibrium between the H_2S gas and the dissolved H_2S at the gas-liquid interface was defined as:

$$[H_2S(i)]_j = k_{H,H_2S}\, P_{H_2S} = k_{H,H_2S}\, y_{H_2S,j}\, P_{tot} \tag{16}$$

The total hydrogen ion concentration in a stage was calculated from the pH of that stage using:

$$[H^+]_j = 10^{-pH_j} \tag{17}$$

2.2.2. Temperature-Dependent Properties

The following temperature-dependent correlations were gathered:

For the equilibrium constant associated with the CO_2 hydration reaction, experimental data were used to develop a well-fitted empirical correlation. The data, originally compiled by Gontse Mokonyane at the University of Pretoria in 2024 [11], were sourced from Soli and Byrne in 2002 [3].

Table 2: Data points for $(1/K_{eq})$ vs Temperature

Temperature (°C)	15.0	17.5	20.0	22.5	25.0	27.5	30.0	32.5
$\boldsymbol{1/K_{eq}}$	841	840	840	843	846	850	866	878

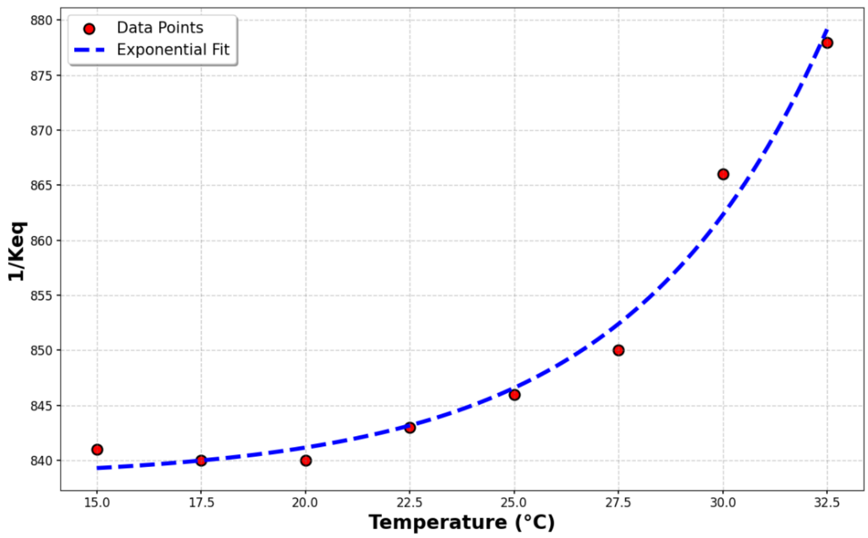

Figure 2: Exponential Fit for $(1/K_{eq})$ vs Temperature Data that was obtained by Mr. Mokonyane [11] Using an Experimental Batch Stripping Process

The exponential fitted relationship for $1/K_{eq}$ as a function of temperature in [°C] was determined to be:

$$\frac{1}{K_{eq}(T)} = 0.040209\, e^{0.213053\,\mathrm{T}} + 838.300799 \tag{18}$$

In Soli & Byrne [3], the first dissociation constant (K_{1C} in [mol/L]) for H_2CO_3 as a function of temperature in [K] (for 15 °C ≤ T ≤ 32.5 °C, at P = 1 atm) is determined as:

$$log K_{1C}(T) = -0.994 - \frac{610.5}{T} \tag{19}$$

In Millero, et al. [7], the second dissociation constant (K_{2C} in [mol/L]) for H_2CO_3 as a function of temperature in [K] (for 0 °C ≤ T ≤ 40 °C, at P = 1 atm) and salinity is determined as:

$$\begin{aligned}-logK_{2C}(T,S) &= pK_{2C}(T,S) \\ &= -452.0940 + 13.142162\,S - (8.101\times10^{-4})S^2 + \frac{21263.61}{T} + 68.483143\,lnT \\ &\quad + \frac{(-581.4428\,S + 0.259601\,S^2)}{T} - 1.967035\,S\,lnT\end{aligned} \tag{20}$$

For this relationship we will take S = 0 to obtain a temperature only dependent relationship for K_{2C}:

$$-logK_{2C}(T) = pK_{2C}(T) = -452.0940 + \frac{21263.61}{T} + 68.483143\,lnT \tag{21}$$

In Millero [8], the first dissociation constant (K_{1S} in [mol/L]) for H_2S as a function of temperature in [K] (for 0 °C ≤ T ≤ 300 °C, at P = 1 atm) is determined as:

$$-logK_{1S}(T) = pK_{1S}(T) = 32.55 + \frac{1519.44}{T} - 15.672\,logT + 0.02722\,T \tag{22}$$

In Stephens & Cobble [9], the second dissociation constant (K_{2S} in [mol/L]) for H_2S as a function of temperature in [K] (for 0 °C ≤ T ≤ 100 °C, at P = 1 atm) is determined as:

$$-logK_{2S}(T) = pK_{2S}(T) = \frac{4500}{T} + 12.6\,log\left(\frac{T}{298.15}\right) - 1.29 \tag{23}$$

In Olofsson & Hepler [10], the ionization constant for water (K_W in [mol^2/L^2]) as a function of temperature in [K] (for 0 °C ≤ T ≤ 300 °C, at P = 1 atm) is determined as:

$$\begin{aligned}-logK_W(T) &= pK_W(T) \\ &= \frac{142613.6}{T} + 4229.195\,logT - 9.7384\,T + 0.0129638\,T^2 \\ &\quad - (1.15068\times10^{-5})T^3 + (4.602\times10^{-9})T^4 - 8909.483\end{aligned} \tag{24}$$

In Soli & Byrne [3], the forward reaction rate constant (k_f in [s^{-1}]) for the rate limiting equilibrium reaction of dissolved CO_2 with water as a function of temperature in [K] (for 15 °C ≤ T ≤ 32.5 °C) is determined as:

$$lnk_f(T) = 22.66 - \frac{7799}{T} \tag{25}$$

Therefore, the first order rate expression for the rate limiting equilibrium reaction of dissolved CO_2 with water would be:

$$r_f = k_f[CO_2(aq)] \tag{26}$$

In Perry's Chemical Engineering Handbook, 9th Edition [6], p.2-280, the dynamic viscosity (μ_L in [Pa.s]) for liquid water as a function of temperature in [K] (at P = 1 atm) is determined as:

$$\mu_L(T) = e^{\left[-52.843+\frac{3703.6}{T}+5.866\, lnT-(5.879\times10^{-29})T^{10}\right]} \tag{27}$$

In Perry's Chemical Engineering Handbook, 9th Edition [6], p.2-99, the molar density (ρ_L in [mol/dm^3] later converted to [kg/m^3]) of liquid water as a function of temperature in [K] (at P = 1 atm) for a temperature range of 273.16 K ≤ T ≤ 353.15 K (0.01 °C ≤ T ≤ 80 °C) is determined as:

$$\rho_L(T) = -13.851 + 0.64038\, T - 0.0019124\, T^2 + 1.8211 \times 10^{-6}\, T^3 \tag{28}$$

In Vargaftik, et al. [13], the surface tension (σ_L in [N/m]) of liquid water as a function of temperature in [K] (at P = 1 atm) is determined as:

$$\sigma_L(T) = (235.8 \times 10^{-3}) \left[\frac{647.15 - T}{647.15}\right]^{1.256} \left[1 - 0.625\left(\frac{647.15 - T}{647.15}\right)\right] \tag{29}$$

Form NIST Webbook, the Antoine coefficients (A, B, C) for the Antoine equation needed to obtain the vapor pressure (P^* in [bar] later converted to [atm]) of water as a function of temperature in [K] (for -17 °C ≤ T ≤ 100 °C) is determined as:

$$A = 4.6543$$

$$B = 1435.264$$

$$C = -64.848$$

$$\log(P^*) = A - \frac{B}{T + C} \tag{30}$$

Form NIST Webbook, the Henry constant for CO_2 (k'_{H,CO_2} in [mol/kg.bar] later converted to [mol/kg.atm]) in water as a function of temperature in [K] (for 0 °C ≤ T ≤ 30 °C, at P = 1 atm) is determined as:

$$k'_{H,CO_2}(T) = 0.034\, e^{2600\left(\frac{1}{T}-\frac{1}{298.15}\right)} \tag{31}$$

Form NIST Webbook, the Henry constant for H_2S (k_{H,H_2S} in [mol/kg.bar] later converted to [mol/kg.atm]) in water as a function of temperature in [K] (for 0 °C ≤ T ≤ 30 °C, at P = 1 atm) is determined as:

$$k_{H,H_2S}(T) = 0.10\, e^{2300\left(\frac{1}{T}-\frac{1}{298.15}\right)} \tag{32}$$

From the 9th Edition of Perry's Chemical Engineering Handbook [6], it is stated that we must use the following expression to determine the effect of temperature on the liquid diffusivity of a gas x in the liquid L:

$$\frac{D_{L,x}(T)\, \mu_L(T)}{T} = \lambda_x = constant \tag{33}$$

In Perry's Chemical Engineering Handbook, 9th Edition [6], p.2-285, we obtain the expression for the liquid diffusivity for CO_2 (D_{L,CO_2} in [cm^2/s] later converted to [m^2/s]) in water as a function of temperature in [K] (at P = 1 atm) as:

$$D_{L,CO_2}(T) = \frac{\lambda_{CO_2}\, T}{\mu_L(T)} \tag{34}$$

where,

$$\lambda_{CO_2} = \frac{D_{L,CO_2}(@25\text{ °C})\ \mu_L(@25\text{ °C})}{(25\text{ °C} + 273.15)} = 6 \times 10^{-15} \tag{35}$$

and,

$$D_{L,CO_2}(@25\text{ °C}) = 1.96 \times 10^{-5}\ cm^2/s \tag{36}$$

In Perry's Chemical Engineering Handbook, 9th Edition [6], p.2-285, we also obtain the expression for the liquid diffusivity for H_2S (D_{L,H_2S} in [cm^2/s] later converted to [m^2/s]) in water as a function of temperature in [K] (at P = 1 atm) as:

$$D_{L,H_2S}(T) = \frac{\lambda_{H_2S}\, T}{\mu_L(T)} \tag{37}$$

where,

$$\lambda_{H_2S} = \frac{D_{L,H_2S}(@25\text{ °C})\ \mu_L(@25\text{ °C})}{(25\text{°C} + 273.15)} = 4.928 \times 10^{-11} \tag{38}$$

and,

$$D_{L,H_2S}(@25\text{ °C}) = 1.61 \times 10^{-5}\ cm^2/s \tag{39}$$

Mass transfer coefficients employed Higbie penetration theory with bubble rise velocities calculated for deformable bubbles (de > 1.3 mm).

The Sherwood number from Nock, et al. [4]:

$$Sh = \frac{k_L\, d_e}{D_L} \Longrightarrow k_L = \frac{Sh\, D_L}{d_e} \tag{40}$$

Where k_L is the liquid-side mass transfer coefficient in [m/s], d_e is the bubble diameter in [m], and D_L is the liquid diffusivity of the gas in the liquid in [m^2/s].

From Nock, et al. [4] the Higbie assumption for a mobile bubble surface moving through a volume of fluid was used:

$$Sh = 1.13\sqrt{Re\,Sc} \tag{41}$$

$$Re = \frac{d_e\, u_b\, \rho_L}{\mu_L} \tag{42}$$

$$Sc = \frac{\mu_L}{\rho_L\, D_L} \tag{43}$$

Where u_b is the rising velocity of the bubble in [m/s], ρ_L is the liquid density in [kg/m^3], μ_L is the dynamic viscosity of the liquid in [Pa.s], and Re and Sc are the dimensionless Reynold's and Scmidt numbers, respectively.

The expression for k_L was found to be:

$$k_L = 2\sqrt{\frac{D_L\, u_b}{\pi\, d_e}} \tag{44}$$

From Guo, et al. [5], an expression for the rising velocity for a bubble in a volume of liquid was found. It was assumed that the bubble diameter will be greater than 1.3 mm and will therefore be of an ellipsoidal shape when it moves through the liquid.

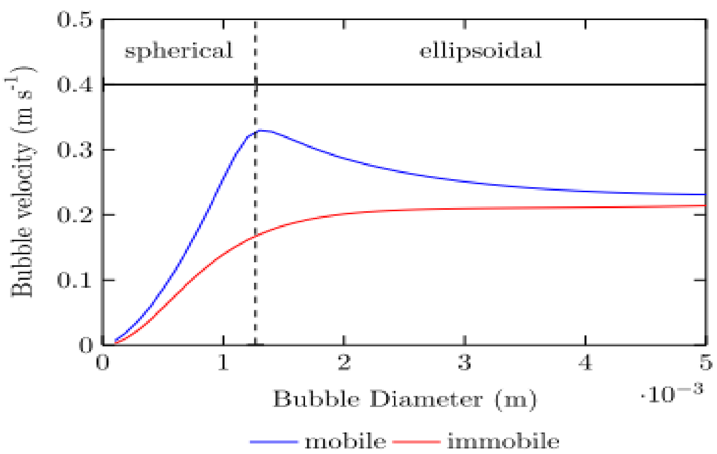

Figure 3: Bubble Rise Velocity vs. Diameter for Mobile and Immobile Bubbles

$$u_b = \sqrt{2.14\frac{\sigma_L}{\rho_L\, d_e} + 0.505\, g\, d_e} \qquad for \quad d_e > 1.3\ mm \tag{45}$$

Where u_b is the rising velocity of the bubble in [m/s], ρ_L is the liquid density in [kg/m^3], d_e is the bubble diameter in [m], σ_L is the surface tension for the liquid in [N/m], and g is the gravitational acceleration constant (which is equal to 9.81 m/s^2).

The expression for the specific interfacial area, a, was determined as follows:

$$a = \frac{total\ interfacial\ area}{total\ volume\ of\ the\ liquid}$$

The units for a will therefore be in [m^{-1}]. For a system with bubbles of droplets, a can be expressed as:

$$a = \frac{6\,\varepsilon}{d_e} \tag{46}$$

Where ε represents the gas-holdup or fraction of the reactor (or tray, for this project) volume occupied by gas. This term is also known as the gas fraction for the reacting system.

An expression for the volumetric mass transfer coefficient was derived in the form of:

$$(k_L a) = 2a\sqrt{\frac{D_L\,u_b}{\pi\,d_e}} \tag{47}$$

Where the volumetric mass transfer coefficient, $k_L a$, has units of [s^{-1}].

2.3. Numerical Solution Strategy

The model solved sequentially from top to bottom in counter-current flow configuration until the specified H_2S recovery target was achieved. Initial conditions at stage 1 were established using the overall H_2S recovery specification:

$$y_{H_2S,1} \cdot G_1 = L \cdot [NaHS]_0 \cdot x_{H_2S} \tag{48}$$

where x_{H_2S} is the fractional recovery target.

For each subsequent stage j, the equation system consisted of 14 unknown variables determined from 14 equations, ensuring a well-posed problem (degrees of freedom = 0). The 14 unknowns comprised:

- Gas flowrate entering from below: G_{j+1}
- Eleven liquid-phase concentrations: $[CO_2(aq)]$, $[H^+]$, $[OH^-]$, $[H_2CO_3]$, $[HCO_3^-]$, $[CO_3^{2-}]$, $[H_2S(aq)]$, $[HS^-]$, $[S^{2-}]$, including the interfacial concentrations $[CO_2(i)]$ and $[H_2S(i)]$
- Two gas-phase mole fractions: $y_{CO_2,j+1}$ and $y_{H_2S,j+1}$

These unknowns were solved using 14 equations consisting of three balance equations (charge, CO_2, H_2S), six equilibrium expressions (K_{1C}, K_{2C}, K_{1S}, K_{2S}, K_W, K_{eq}), two Henry's law relationships, two mass transfer equations, and one vapor composition constraint

Three primary variables were selected to minimize numerical conditioning issues associated with the "stiff" equation system where $[H^+]$ concentrations span many orders of magnitude relative to other species:

- pH (rather than $[H^+]$ directly) - logarithmic transformation
- Scaled carbonic acid concentration (1000×$[H_2CO_3]$)
- Scaled dissolved H_2S concentration (1000×$[H_2S(aq)]$)

The scaling factors of 1000 improved numerical stability by bringing all variables to comparable magnitudes during iterative solution.

For each stage, seven different combinations of initial guesses were systematically tested spanning physically reasonable ranges:

- pH: 4 to 10
- $[H_2CO_3]$: 10^{-6} to 10^{-2} mol/L
- $[H_2S(aq)]$: 10^{-8} to 10^{-1} mol/L

The scipy.optimize.fsolve function employing the hybrid Powell method with tolerance of 10^{-12} solved each nonlinear system. The solution with the lowest sum of squared residuals was selected to ensure global rather than local convergence, avoiding physically unrealistic solutions.

The stopping criterion evaluated whether total sulphur remaining in the liquid phase ($[H_2S(aq)]$ + $[HS^-]$ + $[S^{2-}]$) had decreased below the target value of $[NaHS]_0(1 - x_{H_2S})$, indicating that the specified recovery had been achieved. At this point, further stages would remove more sulphur than available, so the column solution was considered complete, and the total number of stages was recorded. This approach ensures that the model automatically determines the minimum number of stages required for any given recovery specification.

2.4. Simulation Basis and Assumptions

Basis:

- The system was analysed on a per second basis, i.e. liquid and vapor flowrates were in units of [L/s] and [mol/s], respectively.
- The reactive stripping column operates under isobaric conditions (P = 1 atm).
- The reactive stripping column operates under isothermal conditions, and was initially simulated at T = 25 °C.
- The reactive stripping column was initially simulated with the liquid feed stream consisting of 1 L/s of water containing 0.8 mol $NaHS$.
- There was no carbonate present in the liquid feed stream.
- There was no H_2S or S^{2-} present in the liquid feed stream.
- There was no H_2S present in the gas feed stream.
- The CO_2 fed to the column was saturated with water vapour thus y_{H_2O} = 0.0314
- The reactive stripping column was simulated with a gas hold-up of ε = 5 %.
- The reactive stripping column was simulated with a bubble diameter of d_e = 5 mm.
- The reactive stripping column was initially simulated with an outlet vapor flowrate of G_1 = 0.9 mol/s.
- The reactive stripping column was initially simulated with a stage volume of V = 180 L.
- The reactive stripping column was initially simulated with a H_2S recovery of x_{H_2S} = 99.99 % in the outlet vapor stream.

Assumptions:

- The volumetric flow rate of liquid phase flowing down the column was constant and equal to the volume of liquid fed to the column (L = 1 L/s).
- The vapour pressure of water in the vapour stream was constant throughout the column.
- The vapour pressure of water was not affected by changes in the concentrations of the different dissolved species in the liquid stream.

- The sum of the molar concentrations of sulphur compounds i.e. $H_2S(aq)$, HS^-, and S^{2-}, in the liquid feed stream was equal to the molar concentration of Na^+ in the liquid feed stream, and therefore:

$$L([H_2S(aq)]_0 + [HS^-]_0 + [S^{2-}]_0) = L\,[Na^+\,]_0 = L\,[NaHS]_0 \tag{49}$$

$$\therefore [HS^-]_0 = [Na^+\,]_0 = [NaHS]_0 \tag{50}$$

- The equilibrium, dissociation, and ionization constants were not affected by changes in the concentrations of the different dissolved species in the liquid stream.
- The volume of each stage in the reactive stripping column was the same for all stages.
- The gas hold-up for each stage in the reactive stripping column was the same for all stages (ε = 5 %).
- The bubbles rising through the liquid phase in each stage in the reactive stripping column had a constant average diameter (d_e = 5 mm).

3. Software Description

The software implements a comprehensive stage-wise model for reactive H_2S stripping columns, incorporating the critical kinetic limitations of CO_2 hydration that are typically neglected in equilibrium-based approaches. The model solves coupled mass balances, chemical equilibria, and kinetic rate expressions for each stage, providing accurate predictions of column performance across various operating conditions.

3.1. Software Architecture

The software follows a modular architecture with five main components:

- Physical Properties Module: Calculates temperature-dependent properties including Henry's constants, dissociation constants, liquid properties (density, viscosity, surface tension), and mass transfer coefficients.
- Equilibrium Solver: Handles multiple simultaneous chemical equilibria (H_2CO_3 dissociation, H_2S dissociation, water autoionization) and gas-liquid phase equilibria.
- Kinetic Module: Implements CO_2 hydration kinetics using experimentally determined rate constants, accounting for the slow conversion of dissolved CO_2 to H_2CO_3.
- Stage-wise Solver: Performs sequential solution from top to bottom of the column, solving nonlinear equation systems for each stage using scipy.optimize.fsolve.
- Analysis Module: Conducts sensitivity analyses and generates comprehensive visualizations of concentration profiles, pH distribution, and performance metrics.

The architecture ensures numerical stability through variable scaling, multiple initial guess strategies, and robust convergence criteria. Data flow proceeds from input validation through property calculations, stage-wise solution, and finally results analysis and visualization.

3.2. Software Functionalities

Core Modelling Capabilities:

- Stage-wise reactive stripping column simulation
- Integration of CO_2 hydration kinetics with equilibrium processes
- Automatic determination of required stages for target H_2S recovery
- Comprehensive species tracking (8 aqueous species, 3 gas components)

Sensitivity Analysis Tools:

- Operating temperature optimization (15-47°C range)
- Pressure effects analysis (0.3-5 atm)
- Stage volume impact assessment
- Gas flowrate optimization
- Feed concentration effects
- Recovery target analysis

Advanced Features:

- Temperature-dependent physical property correlations
- Mass transfer coefficient calculations with bubble dynamics
- pH profile prediction throughout the column
- Residual analysis for solution validation
- Empirical correlation development for design guidelines

Visualization and Reporting:

- Concentration profile plots for all species
- Gas composition and flowrate tracking
- pH distribution visualization
- Convergence diagnostics and residual analysis
- Automated generation of design charts and correlations

3.3. Sample Code Snippets Analysis

These two code snippets show the core numerical solution approach for each stage in the reactive stripping column model:

```python
# Define equations for this stage with transformed variables
def equations(vars):
    pH, y_scaled, z_scaled = vars  # pH, 1000*c_H2CO3_out, 1000*c_H2S_out

    # Convert back to actual concentrations
    c_H_out = 10**(-pH)
    c_H2CO3_out = y_scaled / 1000.0
    c_H2S_out = z_scaled / 1000.0

    # Calculate other concentrations
    c_OH_out = K_w_val / c_H_out
    c_HCO3_out = c_H2CO3_out * K_1c_val / c_H_out
    c_CO3_out = c_HCO3_out * K_2c_val / c_H_out
    c_HS_out = c_H2S_out * K_1s_val / c_H_out
    c_S_out = c_HS_out * K_2s_val / c_H_out

    c_CO2_out = ((L_flow*(c_H2CO3_out + c_HCO3_out + c_CO3_out) - L_flow*(c_H2CO3_in + c_HCO3_in + c_CO3_in)) /
                 (k_f_val*(1 - ε_gas_fraction/100)*V_stage)) + (c_H2CO3_out/K_eq_val)

    term2 = k_La_CO2*V_stage*(c_CO2_i_stage - c_CO2_out) + k_La_H2S*V_stage*(c_H2S_i_stage - c_H2S_out)
    G_flow_in = G_flow_out + (term2/(1 - y_H2O_out))

    y_H2S_in = ((G_flow_out*y_H2S_out) + (k_La_H2S*V_stage*(c_H2S_i_stage - c_H2S_out))) / G_flow_in
    y_CO2_in = ((G_flow_out*y_CO2_out) + (k_La_CO2*V_stage*(c_CO2_i_stage - c_CO2_out))) / G_flow_in

    eq1 = c_Na_out + c_H_out - c_HCO3_out - 2*c_CO3_out - c_OH_out - c_HS_out - 2*c_S_out
    eq2 = (G_flow_out*y_CO2_out + L_flow*(c_CO2_out + c_H2CO3_out + c_HCO3_out + c_CO3_out) -
           G_flow_in*y_CO2_in - L_flow*(c_CO2_in + c_H2CO3_in + c_HCO3_in + c_CO3_in))
    eq3 = (G_flow_out*y_H2S_out + L_flow*(c_H2S_out + c_HS_out + c_S_out) -
           G_flow_in*y_H2S_in - L_flow*(c_H2S_in + c_HS_in + c_S_in))

    return [eq1, eq2, eq3]
```

Figure 4: Equation System Function

This function defines the three nonlinear equations that must be solved simultaneously for each stage:

- Variable transformation: Uses pH and scaled concentrations (multiplied by 1000) to improve numerical conditioning - this addresses the "stiff" equation problem mentioned where $[H^+]$ is orders of magnitude smaller than other concentrations
- Species calculations: Computes all ionic concentrations from the three primary variables using equilibrium relationships (K_1c_val, K_2c_val, etc.)
- Mass transfer equations: Calculates CO_2 dissolution rate using the kinetic model (k_f_val) and mass transfer coefficients (k_La_CO2, k_La_H2S)
- Three residual equations:
  - eq1: Charge balance (electroneutrality)
  - eq2: CO_2 material balance
  - eq3: H_2S material balance

```python
# Try multiple initial guesses with transformed variables
initial_guesses = [
    [7.0, 1.0, 1.0],      # pH=7, 1000*[H2CO3]=1, 1000*[H2S]=1
    [6.8, 1.2, 1.0],      # pH=6.8, 1000*[H2CO3]=1.2, 1000*[H2S]=1
    [7.2, 0.8, 1.0],      # pH=7.2, 1000*[H2CO3]=0.8, 1000*[H2S]=1
    [7.0, 1.0, 1.2],      # pH=7, 1000*[H2CO3]=1, 1000*[H2S]=1.2
    [7.0, 1.0, 0.8],      # pH=7, 1000*[H2CO3]=1, 1000*[H2S]=0.8
    [6.5, 1.5, 1.5],      # pH=6.5, 1000*[H2CO3]=1.5, 1000*[H2S]=1.5
    [7.5, 0.5, 0.5],      # pH=7.5, 1000*[H2CO3]=0.5, 1000*[H2S]=0.5
]

best_solution = None
best_residual = 1e10
best_equations_residual = None

for guess in initial_guesses:
    try:
        solution = fsolve(equations, guess, xtol=1e-12)
        residual_vals = equations(solution)
        residual = sum([r**2 for r in residual_vals])

        # Check if solution is physically reasonable
        pH_val, y_scaled_val, z_scaled_val = solution
        c_H_val = 10**(-pH_val)
        c_H2CO3_val = y_scaled_val / 1000.0
        c_H2S_val = z_scaled_val / 1000.0

        # Physical constraints
        if (4 <= pH_val <= 14 and c_H2CO3_val > 0 and c_H2S_val > 0 and
            y_scaled_val > 0 and z_scaled_val > 0 and residual < best_residual):
            best_residual = residual
            best_solution = solution
            best_equations_residual = residual_vals
    except:
        continue

if best_solution is None:
    raise ValueError(f"Could not converge stage {stage_num}")
```

Figure 5: Python Solver Strategy

This implements a robust solution approach to handle convergence difficulties:

- Multiple initial guesses: Tests 7 different starting points to avoid local minima and ensure global convergence
- Best solution selection: Chooses the solution with the lowest residual sum-of-squares among all successful attempts
- Physical constraints: Validates solutions by checking pH range (4-14) and positive concentrations
- Error handling: Uses try-except to gracefully handle solver failures and continue testing other initial guesses

This multi-guess strategy is essential because the system involves complex chemical equilibria with wide concentration ranges, making single-point convergence unreliable. The approach ensures that physically meaningful solutions can be found - even when the numerical solver encounters difficulties.

4. Results

4.1. Base Case Simulation

Table 3 shows the input and system variables that were used to obtain the following results from the developed Python program for the base case simulation of the reactive stripping column model:

Table 3: Input and System Variables to for the Base Case Simulation

Variable	Value	Units
$\boldsymbol{T_{operating}}$	25	°C
$\boldsymbol{P_{operating}}$	1	atm
$\boldsymbol{V_{stage}}$	180	L
$\boldsymbol{L}$	1	L/s
$\boldsymbol{G_1}$	0.9	mol/s
$\boldsymbol{\varepsilon}$	5	%
$\boldsymbol{d_e}$	5	mm
$\boldsymbol{[NaHS]_0}$	0.8	mol/L
$\boldsymbol{x_{H_2S}}$	99.99	%

Table 4 shows the resulting values for temperature-dependent equilibrium constants and phase properties, as well as other important parameters.

Table 4: Results for Equilibrium and System Constants

Constant	Value	Units
K_{eq}	1.18123 x10^{-3}	unitless
K_{1C}	9.08600 x10^{-4}	mol/L
K_{2C}	3.85631 x10^{-10}	mol/L
K_{1S}	1.04105 x10^{-7}	mol/L
K_{2S}	1.57371 x10^{-14}	mol/L
K_W	1.00194 x10^{-14}	mol^2/L^2
k_f	0.030259	s^{-1}
a	60	m^{-1}
σ_L	0.071976	N/m
ρ_L	997.0408	kg/m^3
μ_L	9.12531 x10^{-4}	Pa.s
u_b	0.23255	m/s
$k_{L,CO_2}\,a$	0.020441	s^{-1}
$k_{L,H_2S}a$	0.018527	s^{-1}
k'_{H,CO_2}	0.034349	mol/L.atm
k_{H,H_2S}	0.10103	mol/L.atm
P_{vap,H_2O}	0.031378	atm
y_{H_2O}	0.031378	-

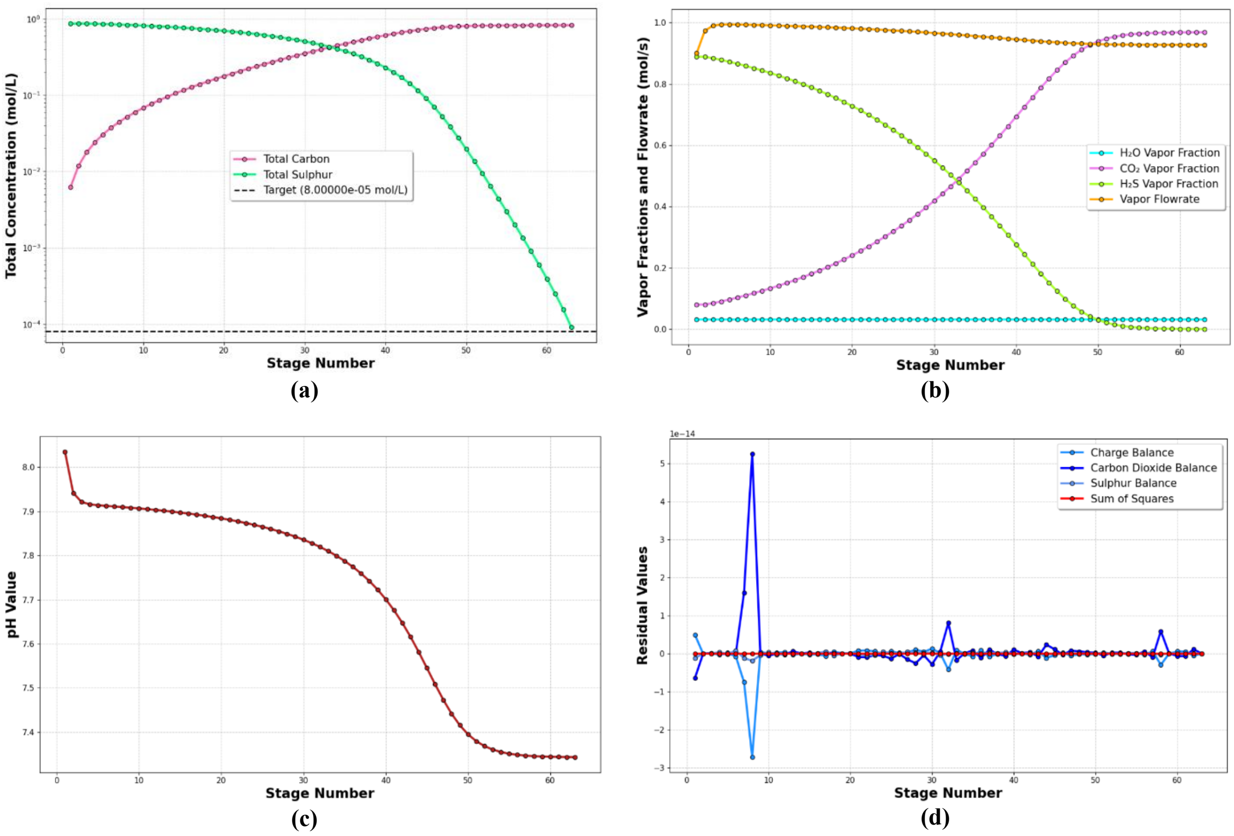

Figure 6: Illustrated Results for the Base Case Simulation Using the Developed Program, **(a)** Total Carbon and Sulphur Species Concentrations, **(b)** Vapor Flowrate and Vapor Fractions, **(c)** pH-Values Throughout the Stripping Column, **(d)** Residual Values Indicating the Accuracy of the Solutions

Concentration profiles in Figure 6(a) demonstrate expected counter-current behaviour: total sulphur decreases from 0.8 to 8×10^{-5} mol/L while total carbon increases from near zero to 0.8 mol/L across 60 stages. Vapor phase composition transitions correspondingly in Figure 6(b), with H_2S fraction declining from 0.9 to near zero and CO_2 fraction rising from 0.08 to 0.95. Shown in Figure 6(c) is the pH gradient (8.05-7.35) which reflects carbonic acid formation consuming alkalinity, creating the driving force for H_2S volatilization.

Residual analysis in Figure 6(d) validates the kinetic model: charge and sulphur balance residuals remain near zero throughout, while CO_2 balance shows a characteristic spike (5×10^{-14}) at stage 8 where kinetic limitations of CO_2 hydration become most pronounced. This confirms that sulphide equilibria are fast relative to the rate-limiting CO_2 hydration step, and the solver successfully captures the interplay between kinetic and equilibrium effects.

4.2. Sensitivity Analysis Simulations

Six operational parameters were systematically varied:

Table 5: Varied Parameters for the Sensitivity Analyses

Operational Parameter	Description	Units	Lower Limit	Upper Limit
$T_{operating}$	Temperature	°C	15	47
$P_{operating}$	Pressure	atm	0.3	5
V_{stage}	Stage Volume	L	50	550
G_1	Gas Flowrate	mol/s	0.825	0.9
$[NaHS]_0$	NaHS Feed Concentration	mol/L	0.5	0.87
x_{H_2S}	H_2S Recovery	%	97	99.999

Empirical correlations were developed using SciPy's curve-fitting algorithms to provide industrial design guidelines.

Table 6: 4th Degree Polynomial Empirical Functions

$y = C_0x^4 + C_1x^3 + C_2x^2 + C_3x + C_4$							
Variable x	**Variable y**	$\boldsymbol{C_0}$	$\boldsymbol{C_1}$	$\boldsymbol{C_2}$	$\boldsymbol{C_3}$	$\boldsymbol{C_4}$	$\boldsymbol{R^2}$
$T_{operating}$ (°C)	pH_N	8.07×10^{-7}	-7.36×10^{-5}	2.55×10^{-3}	-3.23×10^{-2}	7.39×10^{0}	1.0000
$T_{operating}$ (°C)	G_N (mol/s)	-5.22×10^{-8}	5.24×10^{-6}	-1.75×10^{-4}	8.7×10^{-4}	9.53×10^{-1}	0.9999
$T_{operating}$ (°C)	$y_{CO_2,N}$	-6.63×10^{-9}	-4.55×10^{-8}	-2.12×10^{-5}	-3.47×10^{-4}	9.94×10^{-1}	1.0000
V_{stage} (L)	V_{total} (L)	-4.91×10^{-8}	5.85×10^{-5}	-2.14×10^{-2}	1.22×10^{1}	9.46×10^{3}	0.9981
G_1 (mol/s)	$excess\ CO_2$ (%)	-8.51×10^{2}	2.94×10^{3}	-3.8×10^{3}	2.32×10^{3}	-5.72×10^{2}	1.000
$[NaHS]_0$ (mol/L)	pH_N	-4.18×10^{-1}	1.62×10^{0}	-2.61×10^{0}	2.46×10^{0}	6.39×10^{0}	1.0000
$[NaHS]_0$ (mol/L)	G_N (mol/s)	4.85×10^{-3}	-1.31×10^{-2}	2.82×10^{-3}	-6.33×10^{-3}	9.35×10^{-1}	1.0000
x_{H_2S} (%)	$x_{H_2S,actual}$ (%)	8.34×10^{-4}	-3.38×10^{-1}	5.12×10^{1}	-3.44×10^{3}	8.67×10^{4}	1.0000

Table 7: 3rd Degree Polynomial Empirical Functions

$y = C_0x^3 + C_1x^2 + C_2x + C_3$						
Variable x	**Variable y**	$\boldsymbol{C_0}$	$\boldsymbol{C_1}$	$\boldsymbol{C_2}$	$\boldsymbol{C_3}$	$\boldsymbol{R^2}$
x_{H_2S} (%)	pH_N	3.75×10^{-4}	-1.09×10^{-1}	1.06×10^{1}	-3.33×10^{2}	1.0000
x_{H_2S} (%)	$min\ CO_2\ feed$ (mol/L)	-4.77×10^{-5}	1.35×10^{-2}	-1.26×10^{0}	3.95×10^{1}	1.0000

Table 8: Rational Empirical Functions

$y = \dfrac{C_0x^2 + C_1x + C_2}{x^2 + C_3x + C_4}$							
Variable x	**Variable y**	C_0	C_1	C_2	C_3	C_4	R^2
$T_{operating}$ (°C)	N	1.46×10^7	-1.89×10^9	6.93×10^{10}	-2.08×10^7	1.01×10^9	0.9999
$P_{operating}$ (atm)	N	2.53×10^7	1.44×10^8	2.31×10^8	8.08×10^6	-1.49×10^6	0.9989
$P_{operating}$ (atm)	G_N (mol/s)	3.69×10^2	9.6×10^3	-3.75×10^2	1.07×10^4	-3.38×10^22	1.0000
$P_{operating}$ (atm)	$y_{CO_2,N}$	1×10^0	2.3×10^0	-7.3×10^{-2}	2.33×10^0	2.3×10^{-4}	1.0000
V_{stage} (L)	N	1.07×10^4	5.26×10^7	5.82×10^{10}	6.05×10^6	4.37×10^5	1.0000
G_1 (mol/s)	N	4.38×10^1	-6.9×10^1	2.71×10^1	-1.62×10^0	6.56×10^{-1}	0.9996
$excess\ CO_2$ (%)	N	4.39×10^1	6.15×10^1	-9.14×10^2	-4.29×10^0	1.18×10^0	0.9996
$[NaHS]_0$ (mol/L)	N	-5.7×10^0	-7.9×10^1	1.15×10^1	-1.98×10^0	9.67×10^{-1}	0.9998
$[NaHS]_0$ (mol/L)	$min\ CO_2\ feed$ (mol/L)	1.02×10^2	-5.04×10^1	-1.61×10^{-2}	1.02×10^2	-5.05×10^1	1.0000
x_{H_2S} (%)	N	-1.3×10^3	2.44×10^5	-1.13×10^7	-5.15×10^2	4.15×10^4	0.9933

Table 9: Power Law Empirical Functions

$y = C_0x^{C_1} + C_2$					
Variable x	**Variable y**	C_0	C_1	C_2	R^2
$P_{operating}$ (atm)	pH_N	3.14×10^1	-1.41×10^{-2}	-2.41×10^1	1.0000

5. Illustrative Example

The software's capabilities are demonstrated through a comprehensive case study of Na_2CO_3 production from NaHS solution. The base case simulation involves:

Operating Conditions:

- Temperature: 25°C
- Pressure: 1 atm
- Liquid feed: 1 L/s of 0.8 mol/L NaHS solution
- Gas feed: 0.9 mol/s CO_2 (saturated with water vapor)
- Target H_2S recovery: 99.99%
- Stage volume: 180 L each
- Gas holdup: 5%
- Bubble diameter: 5 mm

Key Results:

- Required stages: 63
- Total column volume: 11340 L
- Final pH: 7.35
- Bottom gas composition: 95.2% CO_2, 4.5% H_2O, 0.3% H_2S

The simulation reveals critical insights: the CO_2 hydration reaction (rate constant $k_f = 0.030259\ s^{-1}$) creates significant kinetic limitations, particularly around stage 8 where mass balance residuals spike to 5×10^{-14}, indicating the solver must account for finite reaction rates rather than assuming equilibrium.

Sensitivity Analysis Highlights:

- Optimal operating temperature: 20°C (62 stages minimum), shown in Figure 9(a).
- Optimal pressure: 2.8 atm (38 stages minimum), shown in Figure 9(b).
- Temperature increases to 47°C requires 89 stages due to unfavourable thermodynamics, shown in Figure 9(a).
- Higher NaHS concentrations dramatically increase stage requirements (28 stages at 0.5 mol/L vs 135 stages at 0.87 mol/L)

The software automatically generates empirical correlations for design purposes. For example, the temperature-stage relationship follows:

$$N = \frac{(1.46\times10^{7}T^{2} - 1.89\times10^{9}T + 6.93\times10^{10})}{(T^{2} - 2.08\times10^{7}T + 1.01\times10^{9})}$$

and

$$N = \frac{(2.53\times10^{7}P^{2} - 1.44\times10^{8}P + 2.31\times10^{8})}{(P^{2} - 8.08\times10^{6}P - 1.49\times10^{6})}$$

where N is the number of stages, T is the operating temperature in °C, and P is the operating pressure in atm.

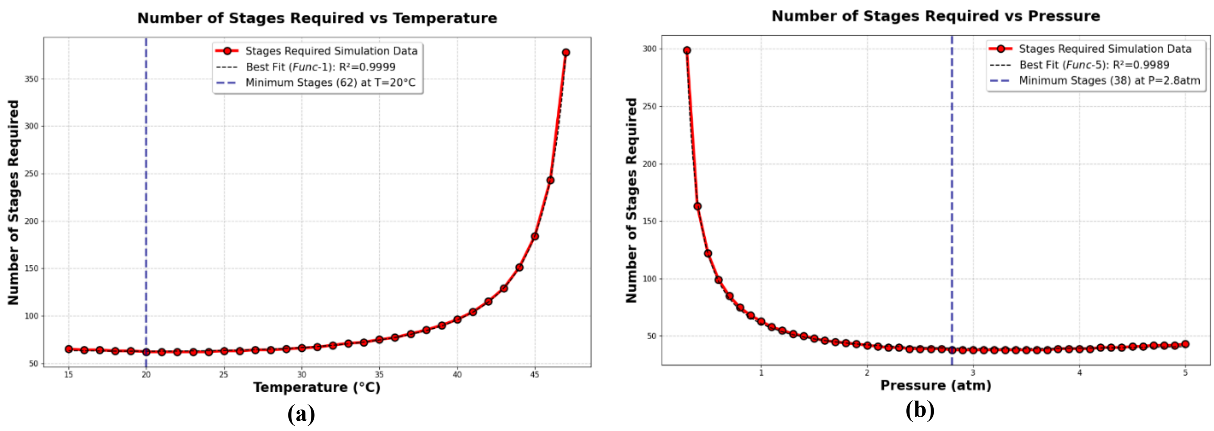

Figure 7: Operating Temperature and Pressure Sensitivity Analyses Highlights, **(a)** Number of Stages Required with Varying Operating Temperature, **(b)** Number of Stages Required with Varying Operating Pressure

6. Impact

New research questions - the software enables investigation of previously inaccessible research areas:

- Quantitative assessment of kinetic vs. equilibrium limitations in reactive mass transfer
- Optimization of multi-ionic systems with competing chemical reaction
- Development of hybrid kinetic-equilibrium models for complex industrial processes
- Investigation of scale-up effects when kinetic limitations dominate

Improved research capabilities - existing research questions are significantly enhanced:

- Process optimization now accounts for realistic kinetic constraints rather than idealized equilibrium assumptions
- Column design incorporates accurate predictions of stage requirements, eliminating over-design or under-performance
- Sensitivity analyses provide quantitative guidance for robust process design
- Integration of multiple time scales (fast equilibria, slow kinetics, mass transfer) in a single framework

Daily practice changes - the software transforms engineering practice by:

- Replacing time-consuming manual calculations with automated comprehensive analysis
- Providing immediate visualization of complex concentration and pH profiles
- Enabling rapid evaluation of multiple design scenarios and operating conditions
- Generating reliable design correlations for industrial implementation

User Adoption: Within the intended user group (chemical engineers working on gas-liquid reactive systems), the software addresses a critical need for kinetic-aware modelling tools. Early adoption by process design engineers and academic researchers has demonstrated its value in avoiding costly design errors that result from neglecting kinetic limitations.

Commercial Impact: The software supports commercial development of sustainable Na_2CO_3 production from waste Na_2SO_4, potentially creating new revenue streams from industrial waste. The modelling framework is adaptable to other reactive stripping applications including CO_2 capture, H_2S treatment in natural gas processing, and pharmaceutical intermediate purification.

Process design companies can use this software to offer more accurate consulting services, while equipment manufacturers benefit from improved column sizing predictions. The quantitative insights enable business cases for waste valorisation projects that might otherwise be considered too risky due to modelling uncertainties.

Scientific Discovery: The software has revealed that kinetic limitations become increasingly dominant at high recovery targets, challenging the conventional wisdom that equilibrium models are adequate for most industrial applications. This finding has implications for many other reactive separation processes and suggests that kinetic considerations should be standard practice in process modelling.

7. Conclusions

This software fills a critical gap in reactive stripping column modelling by incorporating CO_2 hydration kinetics that are typically neglected in equilibrium-based approaches. The comprehensive Python implementation provides process engineers with accurate tools for column design and optimization, while the extensive sensitivity analysis capabilities enable robust process development.

Key contributions include:

- Integration of slow CO_2 hydration kinetics with fast chemical equilibria in a numerically stable framework
- Demonstration that kinetic limitations become dominant at high recovery targets
- Identification of optimal operating conditions that balance kinetic and thermodynamic effects
- Development of empirical design correlations for industrial application.

The software enables more sustainable chemical processing by accurately modelling waste valorisation processes, supports development of new separation technologies, and provides a framework for investigating kinetic limitations in other reactive systems. Future developments will include experimental validation, extension to non-isothermal operations, and integration with process optimization algorithms.

Conflict of Interest

No conflict of interest exists: We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

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References

[1] ChemicalBook, 2022. Uses of Sodium Carbonate. [Online] Available at: https://www.chemicalbook.com/article/uses-of-sodium-carbonate.htm [Accessed 10 April 2025].

[2] Dreybrodt, W., Lauckner, J., Zaihua, L., Svensson, U. and Buhmann, D., 1996. The kinetics of the reaction $CO_2 + H_2O \rightarrow H^+ + HCO_3^-$ as one of the rate limiting steps for the dissolution of calcite in the system H_2O-CO_2-$CaCO_3$. Geochimica et Cosmochimica Acta, 60(18), pp. 3375-3381.

[3] Soli, A.L. and Byrne, R.H., 2002. CO_2 system hydration and dehydration kinetics and the equilibrium CO_2/H_2CO_3 ratio in aqueous NaCl solution. Marine Chemistry, 78(2-3), pp. 65-73.

[4] Nock, W., Heaven, S. and Banks, C., 2015. Mass transfer and gas–liquid interface properties of single CO_2 bubbles rising in tap water. Chemical Engineering Science, 140, pp. 171-178.

[5] Guo, X., Silaen, A.K., Walla, N.J., Oltmann, H.G. and Zhou, C.Q., 2018. The mechanism of inclusion removal from molten steel by dissolved gas flotation. Ironmaking & Steelmaking, 45(7), pp. 648-654.

[6] Perry, R.H. and Green, D.W. (eds.), 2018. Perry's Chemical Engineering Handbook, 9th ed. New York: McGraw-Hill Education.

[7] Millero, F.J., Graham, T.B., Huang, F., Bustos-Serrano, H. and Pierrot, D., 2002. Dissociation constants for carbonic acid determined from field measurements. Deep-Sea Research Part I, 49(10), pp. 1705-1723.

[8] Millero, F.J., 2003. The thermodynamics and kinetics of the hydrogen sulfide system in natural waters. Marine Chemistry, 83(3-4), pp. 209-227.

[9] Stephens, H. and Cobble, J.W., 1970. Thermodynamic Properties of the Aqueous Sulfide and Bisulfide Ions and the Second Ionization Constant of Hydrogen Sulfide over Extended Temperatures. Inorganic Chemistry, 9(2), pp. 425-428.

[10] Olofsson, G. and Hepler, L.G., 1975. Thermodynamics of ionization of water over wide ranges of temperature and pressure. Journal of Solution Chemistry, 4(2), pp. 127-143.

[11] Mokonyane, G., 2024. Stripping Hydrogen Sulphide from an Aqueous Solution of Sodium Hydrosulphide. BEng Final Year Project Report, University of Pretoria, Pretoria.

[12] Vargaftik, N.B., Volkov, B.N. and Voljak, L.D., 1983. International Tables of the Surface Tension of Water. Journal of Physical and Chemical Reference Data, 12(3), pp. 817-820.

[13] Schulz, K.G., Riebesell, U., Rost, B., Thoms, S. and Zeebe, R.E., 2006. Determination of the rate constants for the carbon dioxide to bicarbonate inter-conversion in pH-buffered seawater systems. Marine Chemistry, 100(1-2), pp. 53-65.

[14] van Tonder, D.A., 2025. Reactive H_2S Stripping Column Model Software Repository. Zenodo. DOI: 10.5281/zenodo.17210266

Current executable software version

Table 10: Software Metadata

Nr	Executable software metadata description	Executable software metadata
S1	Current software version	2.0.1
S2	Permanent link to executables of this version	https://github.com/DamianvanTonder/Reactive_H2S_Stripping_Column_Simulation_Model_with_CO2_Hydration_Kinetics/releases/tag/v2.0.1
S3	Legal Software License	MIT Licence
S4	Computing platforms/Operating Systems	• Linux • OS X • Microsoft Windows
S5	Installation requirements & dependencies	• Python ≥ 3.8.0 • NumPy ≥ 1.20.0 • SciPy ≥ 1.7.0 • Pandas ≥ 1.3.0 • Matplotlib ≥ 3.3.0
S6	If available, link to user manual - if formally published include a reference to the publication in the reference list	N/A
S7	Support email for questions	u21449300@tuks.co.za